The Torres Lab in one page

The problem

About 2.7 billion people cook and heat with wood and other biomass; wood smoke is the leading COPD risk factor among them, and wildfire smoke now brings the same exposure to cities. The damage starts in the small airways, where repeated injury turns a repair response into permanent scarring (subepithelial fibrosis).

The biology chain we model

Smoke lands on the airway lining -> epithelial cells are injured and release cytokines -> macrophages respond, first a 'fight' state (M1) then a 'repair' state (M2) -> repeated exposure keeps the repair signal (TGF-beta) on -> fibroblasts become myofibroblasts -> collagen accumulates under the lining, the wall thickens, the airway narrows.

Chapter 1 (2019-2024): equations

Nine quantities (healthy and damaged epithelium, macrophage states, two cytokine pools) and nine rates of change, each term one arrow in the chain above, each rate a number with units from cell-culture measurements.
Bakry, Brashear, Brashear, Jones & Torres (2024), Journal of Theoretical Biology.

Chapter 2 (2025-now): a virtual tissue

A 250 x 125 micrometre slice of airway wall in CompuCell3D: cells are clusters of pixels that push, stick and follow chemical gradients; the equations from Chapter 1 run inside and set the injury. 1 pixel = 1.25 um, 1 step = 62.4 s, 20,000 steps = 14 days. Every parameter is a published measurement or labelled 'free'.

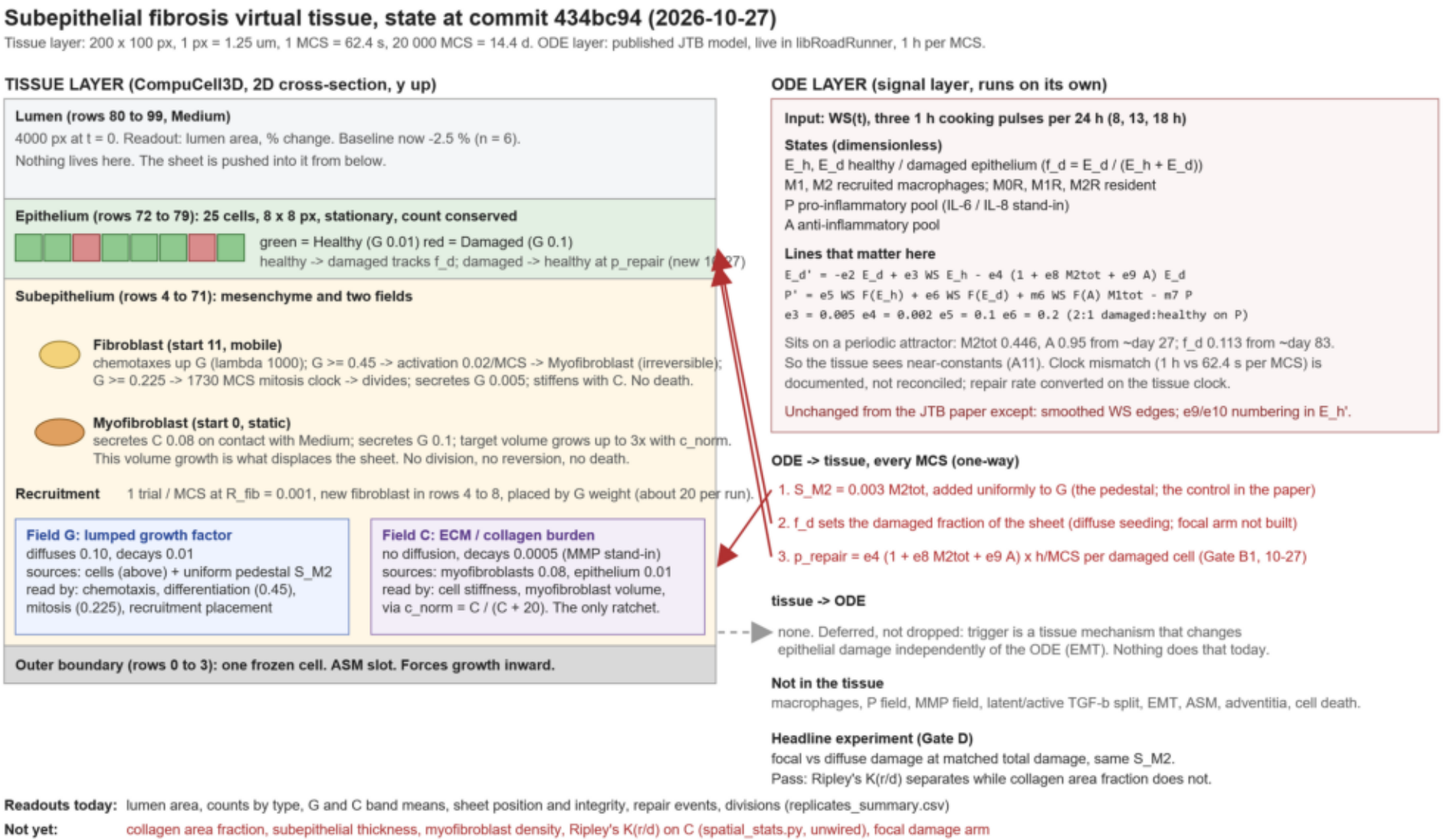
The question for the 2027 paper

Same total injury, different arrangement (one patch vs scattered cells): does the scar come out different?
The test: Ripley's K on the collagen pattern separates focal from diffuse while total collagen does not.

What a student does here

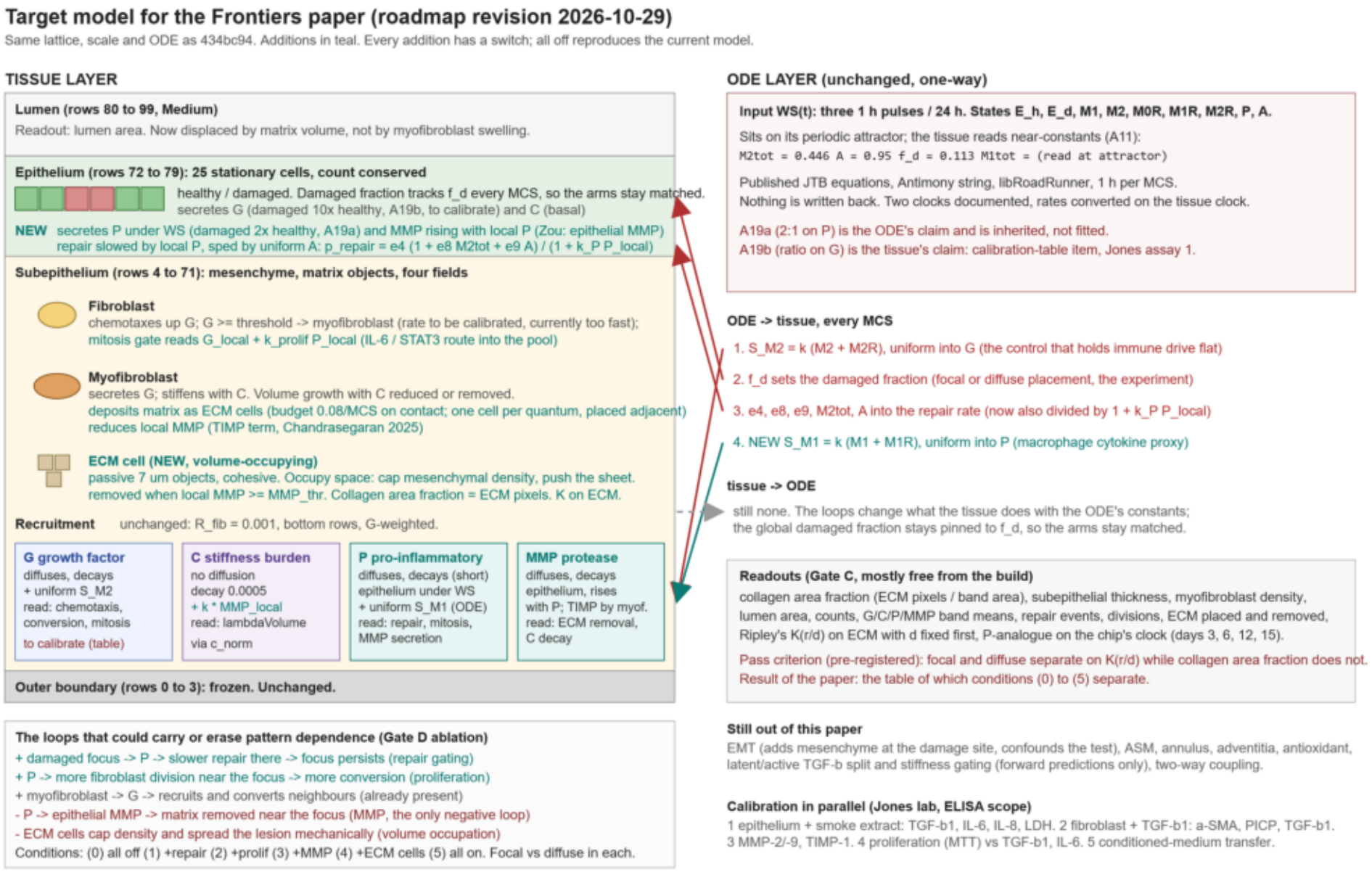
Turn a paper into a number with units and a confidence. Decide which of two conflicting studies the model should trust. Run the equations in Tellurium, change one parameter, explain what happened in biological words.
Design the ELISA that would pin a free parameter. Present a poster. Ask of every claim: compared to what, measured how?

The virtual tissue as it runs today (state at 2026-10-27)



Left: the tissue layer, four bands from the airway lumen (top) to the fixed outer wall (bottom), with the two chemical fields. Right: the ordinary-differential-equation layer (the published immune-response model) and the three quantities it hands to the tissue every step. Nothing is written back: the tissue reads the immune model as a set of near-constant inputs.

The model the 2027 paper will use (roadmap revision 2026-10-29)

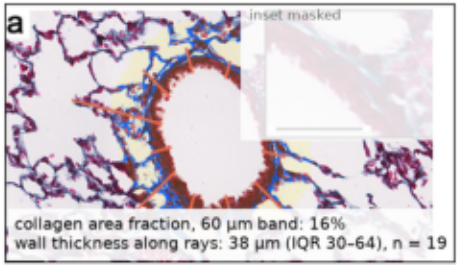


Additions in teal, each behind a switch: a spatial pro-inflammatory field with three consumers, an MMP (collagen-digesting enzyme) field, and collagen as volume-occupying objects. The focal-vs-diffuse experiment is run with each loop off and on, so the result is a table of which mechanisms carry or erase pattern dependence.

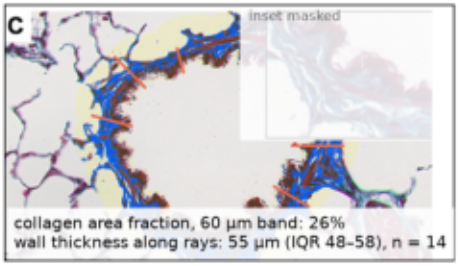
Proposed analysis: the same measurements on a real section and on the virtual tissue

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Rat small airway, control (7 months)
Masson's trichrome; Zou et al. 2014 Fig 4, CC BY 4.0



Rat small airway, wood smoke (7 months)
Masson's trichrome; Zou et al. 2014 Fig 4, CC BY 4.0



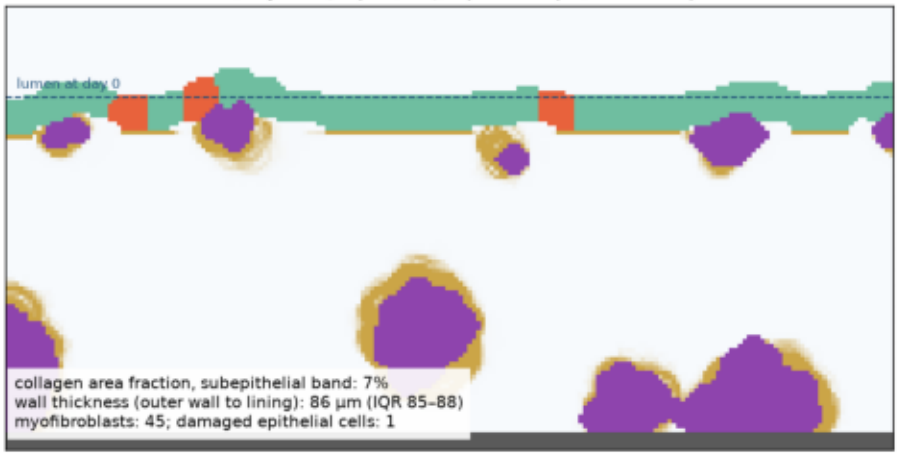
How the overlays are made

1. Find the airway lumen: the large bright region in the middle of the field.
2. Draw a 60 µm band around it (pale yellow): that is the airway wall we score.
3. Collagen = trichrome blue pixels inside the band (blue overlay). Area fraction = blue / band.
4. Wall thickness = distance from the lumen edge to the first open air space, along rays (orange lines). Report the median and IQR.
5. Scale from the 50 µm bar in each panel.

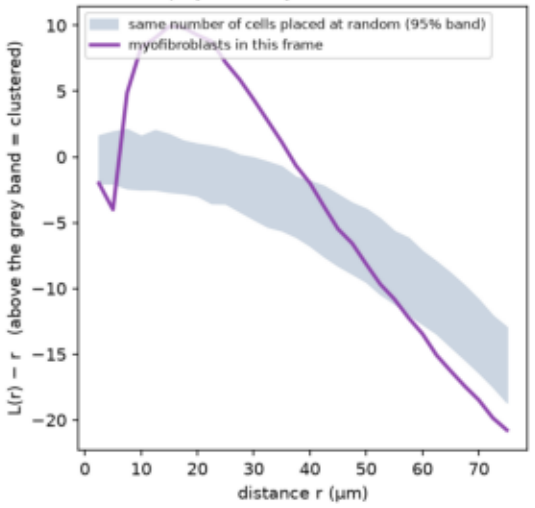
The same four numbers come out of the model (below) by counting pixels, so model and tissue can be compared on equal terms.

These overlays are a demonstration of the method on a published figure, not a re-analysis of the study; Zou et al. report their own morphometry.

Virtual tissue, day 14.4 (step 20,000); 1 px = 1.25 µm, 250 × 125 µm shown

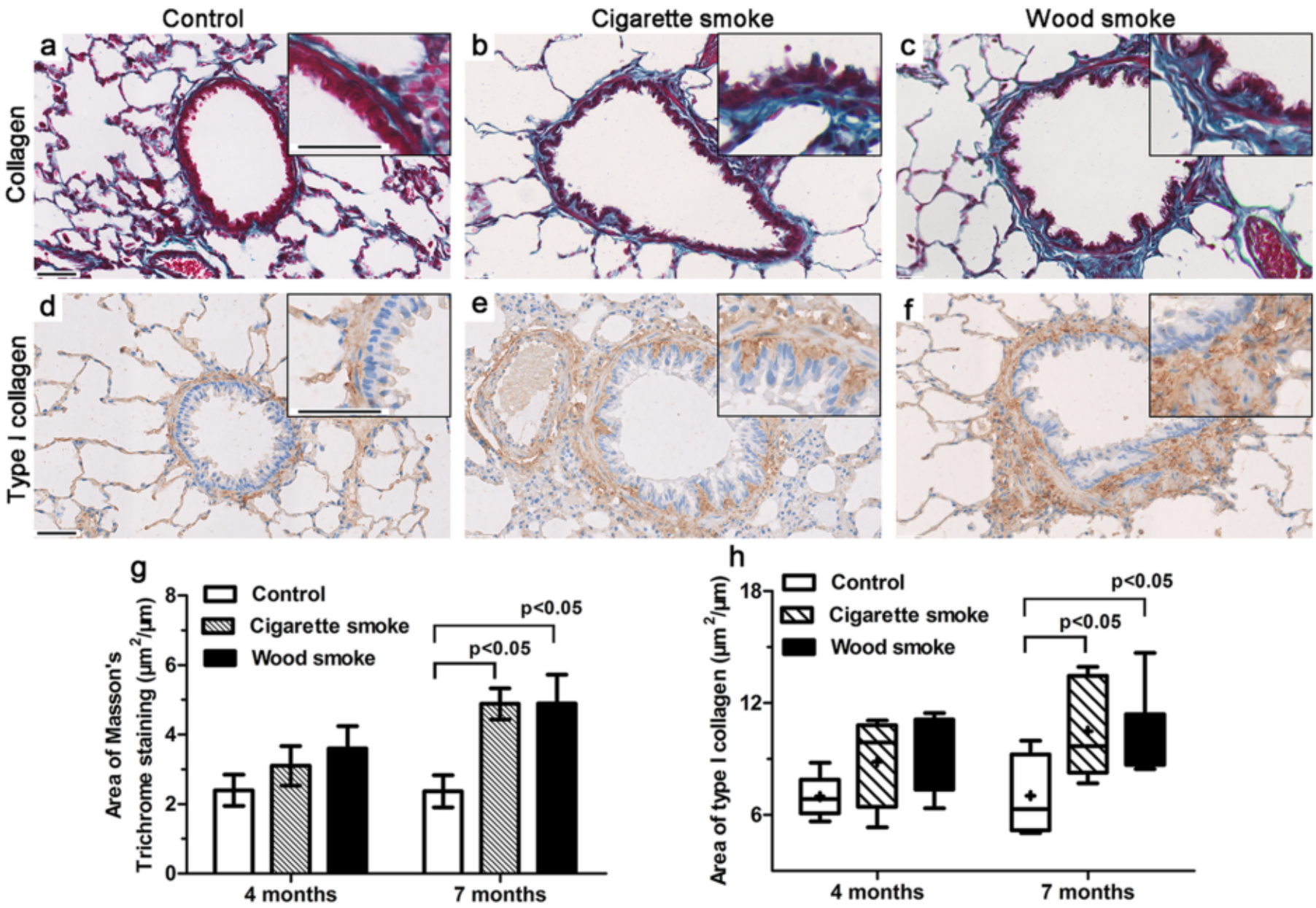


Ripley's K on myofibroblast centres



Top: Masson's trichrome sections of rat small airways after 7 months (Zou et al. 2014, PLoS ONE 9:e96708, Fig. 4, CC BY 4.0) with a 60 um band around the lumen, trichrome-blue collagen in the band (collagen area fraction) and wall thickness along rays. Bottom: the same two readouts on the final model frame, and Ripley's K on myofibroblast centres against random placement. The overlays demonstrate the method on a published figure; they are not a re-analysis of that study.

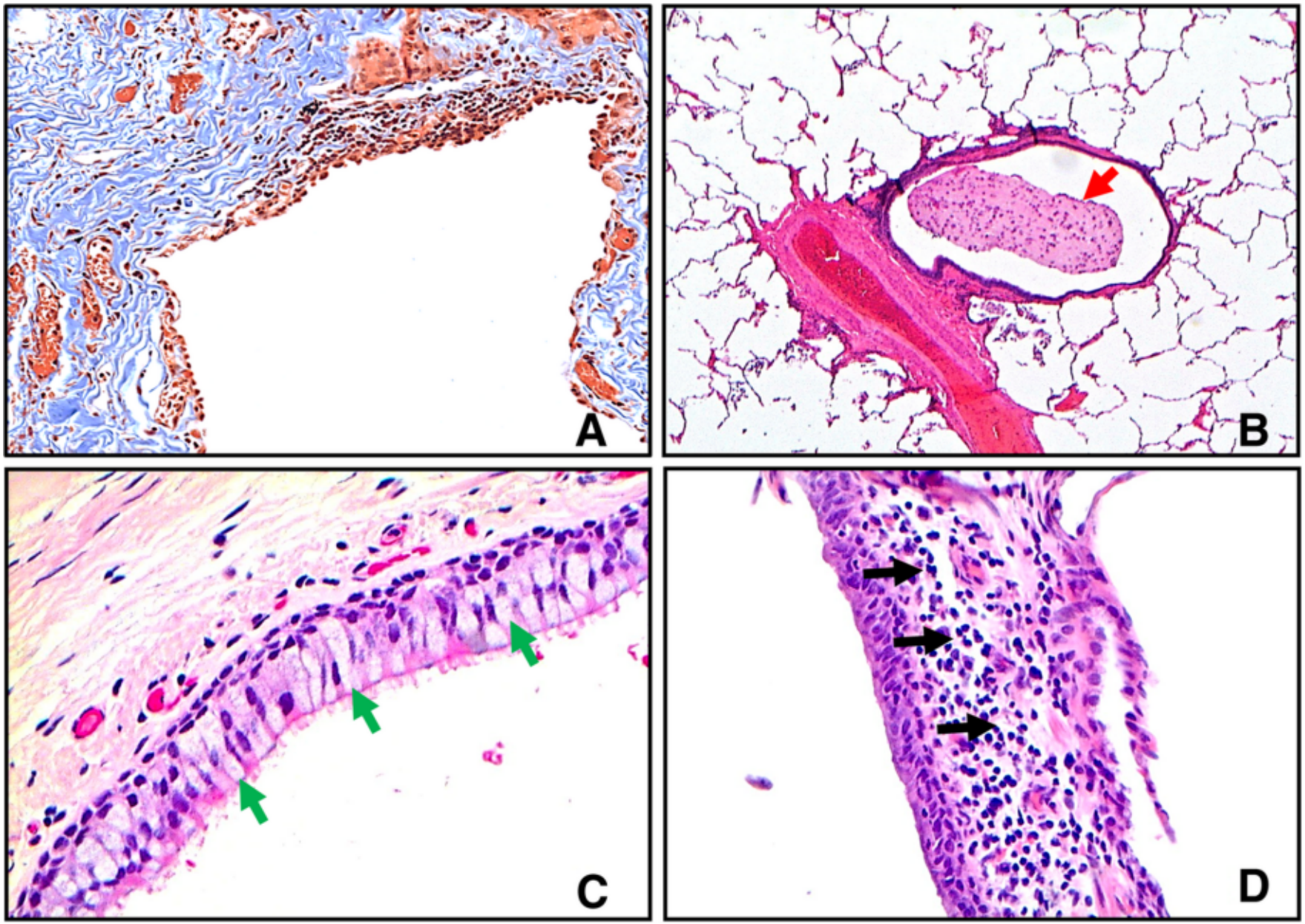
Real tissue: collagen builds up around small airways after months of smoke



Rat small airways after 7 months of exposure: control, cigarette smoke, wood smoke. Top row Masson's trichrome (collagen blue); bottom row immunostaining for type I collagen (brown); (g, h) quantification. This is the blue ring the model's amber 'collagen burden' stands for.

Zou et al. 2014, PLoS ONE 9:e96708, Fig. 4, doi:10.1371/journal.pone.0096708 (CC BY 4.0)

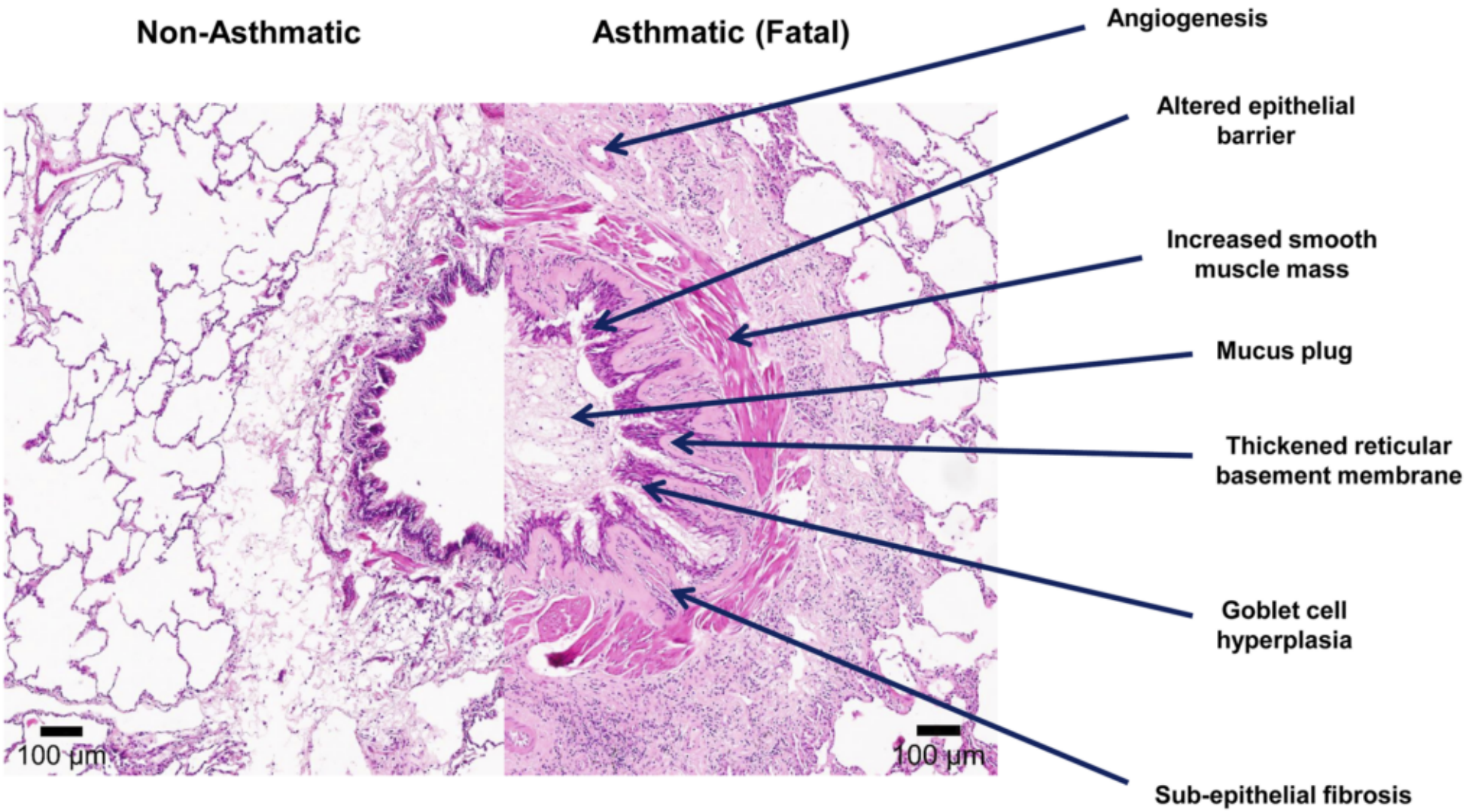
Real tissue: human COPD small airways



(A) Masson's trichrome: a COPD bronchiole with a thick fibrotic wall of blue collagen bundles under the epithelium. (B) H&E: a mucus plug in the lumen. (C) goblet-cell hyperplasia. (D) inflammatory cells in the wall. Human COPD small airways narrow by roughly 22%; the model currently reaches about 2.5%, a gap we list as a limitation and a question.

Higham, Quinn, Cancado & Singh 2019, Respir Res 20:49, Fig. 2, doi:10.1186/s12931-019-1017-y (CC BY 4.0)

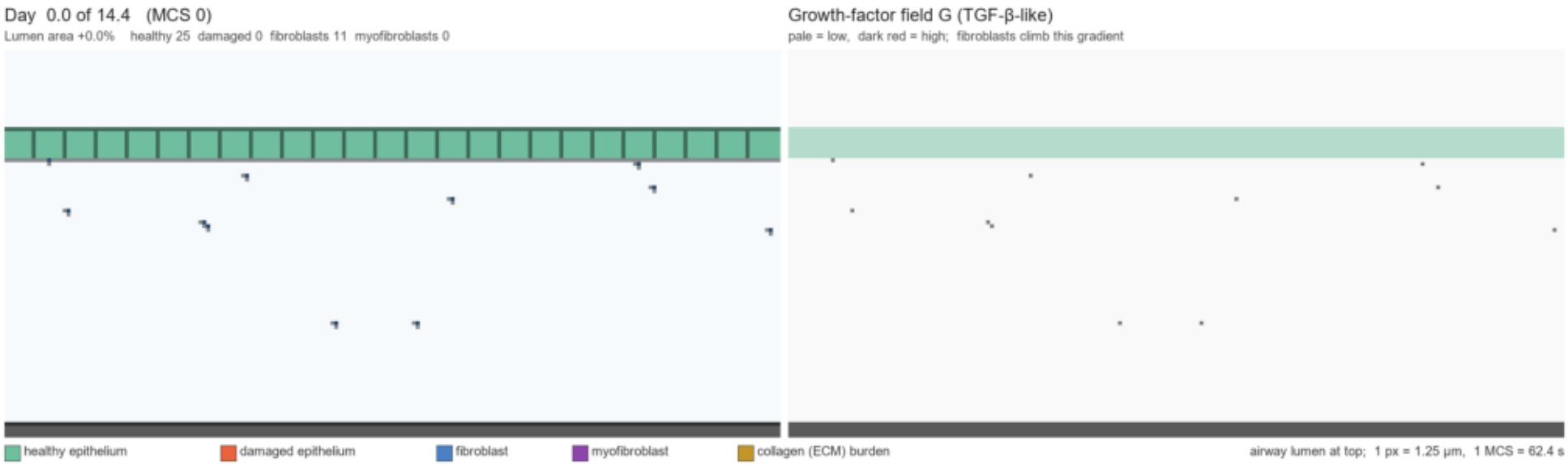
Real tissue: a labelled section showing sub-epithelial fibrosis (asthma)



H&E of a non-asthmatic and a fatal-asthma human airway. Not a smoke lesion, but the clearest labelled picture of the layer the model simulates: the arrow marked sub-epithelial fibrosis.

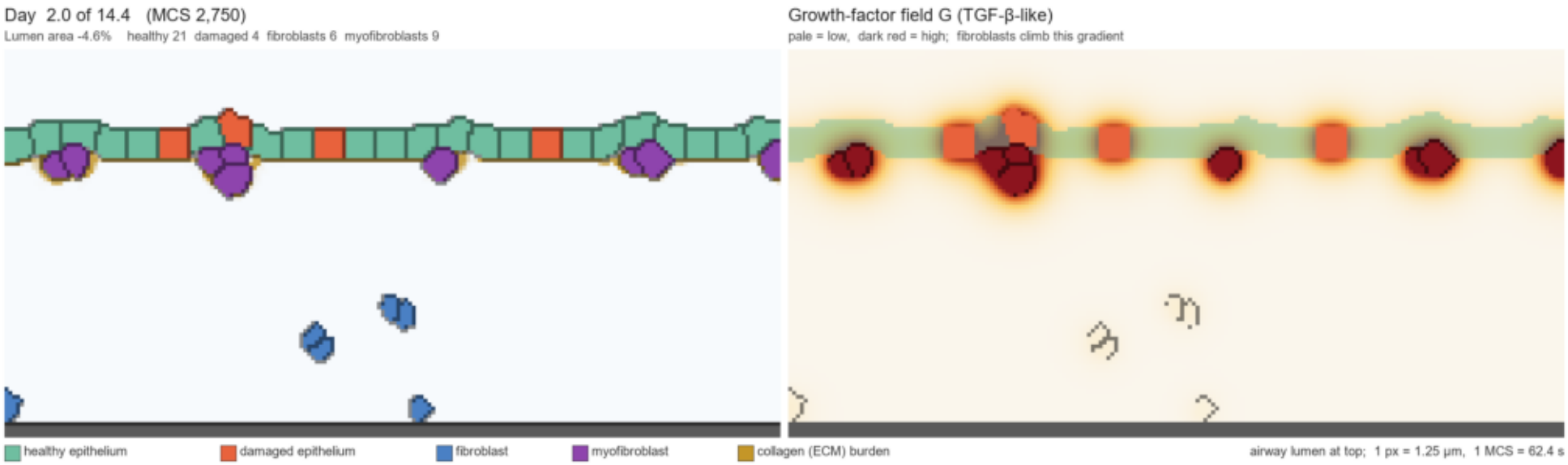
Hsieh, Assadinia & Hackett 2023, Front Physiol 14:1113100, Fig. 1, doi:10.3389/fphys.2023.1113100 (CC BY 4.0)

Model frame: day 0



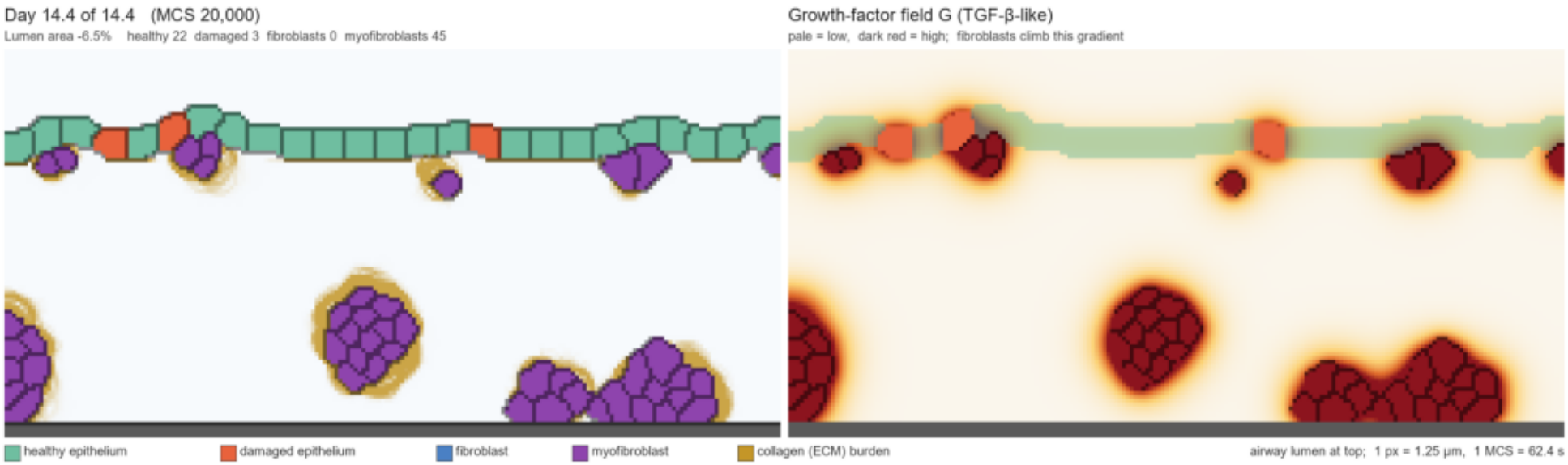
Left: cells (green healthy epithelium, orange damaged, blue fibroblasts, purple myofibroblasts, amber collagen burden). Right: the growth-factor field the fibroblasts climb. Lumen at top; 1 px = 1.25 um; 1 step = 62.4 s.

Model frame: day 2



Left: cells (green healthy epithelium, orange damaged, blue fibroblasts, purple myofibroblasts, amber collagen burden). Right: the growth-factor field the fibroblasts climb. Lumen at top; 1 px = 1.25 um; 1 step = 62.4 s.

Model frame: day 14



Left: cells (green healthy epithelium, orange damaged, blue fibroblasts, purple myofibroblasts, amber collagen burden). Right: the growth-factor field the fibroblasts climb. Lumen at top; 1 px = 1.25 um; 1 step = 62.4 s.